Questions

Q1.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which of these changes of state describes sublimation?

(1)

- ☐ **A** from gas to liquid
- ☐ **B** from liquid to solid
- ☐ **C** from solid to gas
- ☐ **D** from solid to liquid

(Total for question = 1 mark)

Q2.

Figure 1 shows some gas in a container.

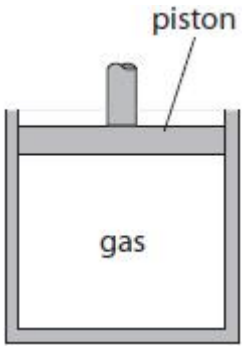

Figure 1

The piston is forced down to a new position, compressing the gas, as in Figure 2.

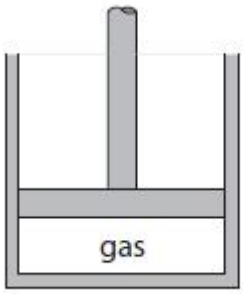

Figure 2

The temperature of the gas remains constant.
Explain, in terms of particles, why the pressure of the gas increases.

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(Total for question = 2 marks)

Q3.

Figure 5 shows some water in a measuring cylinder and a lump of iron.

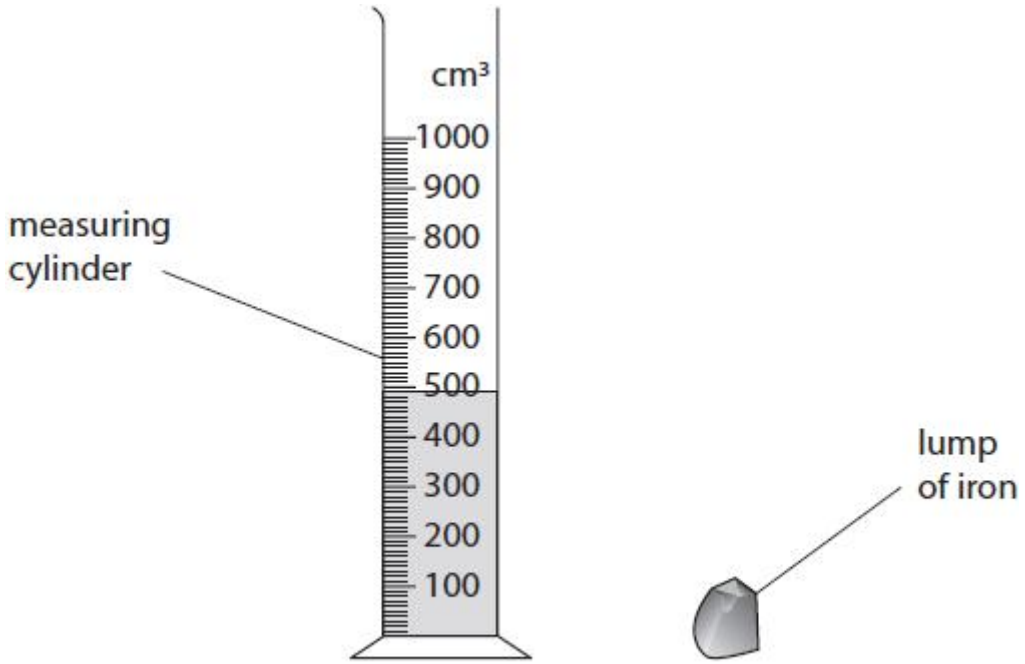

Figure 5

The lump of iron is lowered fully into the water.

The water level in the measuring cylinder rises to 530 cm^3.

The density of iron is 7.9 g / cm^3.

A piece of wood has a similar shape and volume to the lump of iron.

The density of the wood is 0.82 g / cm^3.

The density of water is 1.00 g / cm^3

Explain why the previous method used cannot be used to determine the mass of the piece of wood.

(2)

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(Total for question = 2 marks)

Q4.

A container of gas is at room temperature.

The gas is then heated.

The volume of the container remains the same.

By considering changes in velocities of the gas particles, explain how the temperature increase affects

- the average kinetic energy of the particles
- the pressure the particles exert on the walls of the container.

(6)

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(Total for question = 6 marks)

Q5.

Figure 7 gives information about the density of aluminium.

	density in g/cm^3
solid aluminium	2.70
liquid aluminium	2.38

Figure 7

Explain the difference between the density of solid aluminium and the density of liquid aluminium in terms of the arrangement of particles.

(2)

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(Total for question = 2 marks)

Q6.

Explain the difference between the term 'specific heat capacity' and the term 'specific latent heat' when applied to heating substances.

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(Total for question = 2 marks)

Q7.

Figure 19 shows some apparatus that may be used to determine the specific heat capacity of water.

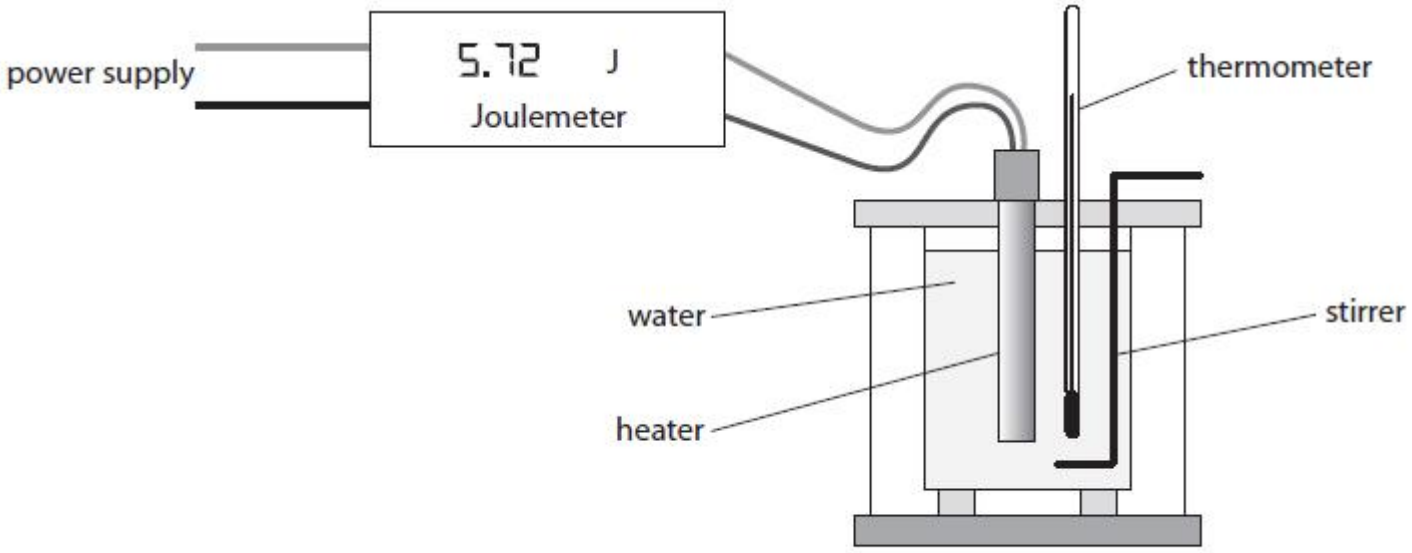

Figure 19

A student measures the initial temperature of the water.

The power supply is switched on for 10 minutes and then switched off.

Explain how the student should then obtain an accurate reading for the final temperature of the water, to be used in the calculation of the specific heat capacity.

(3)

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(Total for question = 3 marks)

Q8.

Figure 1 shows some gas in a container.

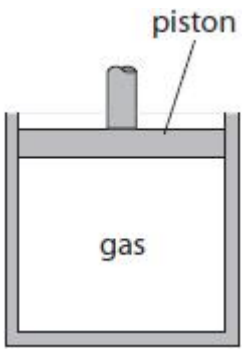

Figure 1

Explain, in terms of particles, how the gas exerts a pressure on the piston.

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(Total for question = 2 marks)

Q9.

Describe what happens when a substance experiences sublimation.

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(Total for question = 2 marks)

Mark Scheme

Q1.

Question Number	Answer	Mark
	The only correct answer is **C** from solid to gas ***A*** *is 'condensation'* ***B*** *is 'freezing'* ***D*** *is 'melting'*	**(1)** **AO1.1**

Q2.

Question number	Answer	Mark
	An explanation that combines identification – knowledge and reasoning / justification • volume decrease makes the density of particles increase / more crowded idea (1) • increasing the **rate** at which particles collide (with the piston) (1)	**(2)**

Q3.

Question number	Answer	Additional guidance	Mark
	an explanation linking density of wood less (than that of water) (1) less (volume of) water displaced (than volume of wood) (1)	 allow wood floats / should be submerged allow wood absorbing water allow (idea of) incorrect volume reading allow (idea that) the volume cannot be measured this way	**(2)** **AO2.2**

Q4.

SSQ NO:	CS NO:	Answer	Mark
*		Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO1 strand 1 (6 marks)** • particles move faster (at a higher temperature) • greater velocity / speed means greater kinetic energy • since KE = ½ m v^2 • heating increases KE (store) • KE (store) increase leads to higher (average) speeds • faster particles (at higher temperature so) hit container with more force / momentum exchange • bigger pressure because p = F / A • particles hit container more frequently (at higher temperature) • so **more** force exerted on (walls of) container	**(6)** **AO1.1**

Level	Mark	Descriptor
	0	No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

Summary for guidance

Level	Mark	Additional Guidance	General additional guidance – the decision within levels Eg - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance isolated idea(s) of physics e.g. recognising the speed-temperature relationship or the pressure temperature relationship	Possible candidate responses particles faster (at higher temperature) KE increases pressure increases (at a higher temperature)

Level 2	3–4	Additional guidance limited details about KE **or** limited details about pressure **or** linked ideas about kinetic energy and pressure	Possible candidate responses faster particles have greater kinetic energy (store) (particles) hitting container more often causes greater pressure faster particles cause greater force bigger pressure because force increased
Level 3	5–6	Additional guidance understanding is detailed and fully developed. includes detail about **both** kinetic energy **and** force involvement in pressure, but one aspect may be covered in greater detail than the other one	Possible candidate responses greater speed means greater kinetic energy since KE = ½ m v^2 **AND** bigger pressure because more frequent collisions causes an increase in force greater speed means greater kinetic energy **AND** bigger pressure because p = F / A and (total) force increased because of hitting container walls with bigger momentum (changes)

Q5.

	Answer	Additional guidance	Mark
	an explanation linking: density of solid is greater (than density of liquid) (1) (because) distance between particles in solid is less (than distance between particles in liquid) (1)	 solids are denser accept in solids, particles are closer accept in solids, there are **more** particles per unit volume / particles are **more** (tightly) packed	**(2)** **AO1.1**

Q6.

Question number	Answer	Additional guidance	Mark
	an explanation linking specific heat capacity concerns change in temperature (1) whereas specific latent heat concerns change of state (1)	 accept specific heat capacity concerns heating <u>up</u> / cooling accept any named change of state e.g. melting / freezing / evaporating /boiling accept specific latent heat related to no change in temperature	**(2)** **AO1.1**

Q7.

Question number	Answer	Additional guidance	Mark
	an explanation linking any **three** from: stir the water before taking a reading of temperature (1) (continue to) observe temperatures after switching off (1) record the maximum / highest / peak temperature reached (1) take temperature reading at eye level (1) conduction (and convection) take time (1)	 allow "for **longer** than 10 minutes" allow wait(ing period) in correct context until the temperature stops changing takes time (for water / thermometer) to heat through	**(3)** **AO1.2**

Q8.

Question number	Answer	Mark
	An explanation that combines identification –knowledge and reasoning / justification • (particles / atoms / molecules) {hit / collide with} piston (1) • causing a force (on the piston) (1)	**(2)**

Q9.

Question number	Answer	Additional guidance	Mark
	A description including idea of change of state / solid **changes** (1) to gas / vapour (directly) (1)	 accept equivalents e.g. turns into / goes from to allow reverse i.e. gas → solid may be via appropriate example e.g. ice → water vapour / steam or reverse (2 marks)	**(2)** **AO1.1**